Continuous Compounding of Green/Sustainable Finance Research without Structural Breaks: A Bibliometric Study with Endogenous Break Detection

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Abstract

This study examines the publication dynamics of green and sustainable finance research within the banking literature using a large-scale bibliometric framework. Drawing on 498,958 banking-domain articles from Semantic Scholar (2000–2025), we identify 4,901 papers classified by 15 word-boundary stems and apply endogenous structural break detection to the yearly publication series. Unlike technology-driven waves such as AI and blockchain, the green finance wave exhibits no significant structural break at any candidate date: all Chow tests are null (2017 $p = 0.26$, 2020 $p = 0.13$, 2022 $p = 0.11$), and the endogenous supremum F-test maximum is only $F = 2.0$ (n.s.). Instead, green finance grows continuously from near-zero in 2015 to 1,760 papers in 2025, reaching 5.1% of all banking research output. Mann-Kendall $z = 6.63$ ($p < 0.001$) with Sen's slope of 0.30 confirms the monotonic upward trend. A manual audit of 100 era-stratified records yields a pooled precision of 91% (Wilson CI: 84–95%), stable across eras (92% pre-2017, 90% 2017–2021, 90% post-2022), representing the cleanest classification of all three technology waves studied. Venue tier analysis shows 95–100% of green finance papers appear in established venues across all eras. Cross-wave correlation analysis reveals that the green finance wave moves independently of both AI ($r = 0.076$, $p = 0.71$) and blockchain ($r = 0.104$, $p = 0.61$). These findings demonstrate that green finance diffusion in banking follows a continuous, policy-driven compounding trajectory rather than a technology shock, distinguishing it from capability-driven waves and providing methodological contributions to the structural break analysis of bibliometric data.

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1. Introduction

Green and sustainable finance has emerged as one of the most rapidly growing research domains within banking and financial studies. From near-zero publication output before 2015, the field has expanded to 1,760 papers in 2025, constituting 5.1% of all banking-domain research. This growth trajectory coincides with a series of landmark international policy events—the Paris Agreement (2015), the UN Sustainable Development Goals (2015), and COP26 in Glasgow (2021)—that have fundamentally reshaped the research agenda of financial institutions and the academic community studying them.

Despite this explosive growth, several fundamental questions remain unanswered. First, has green finance research in banking experienced genuine structural breaks in its publication trajectory—that is, discrete shifts in the underlying growth regime—or has growth been continuous? This question has direct implications for how we understand the relationship between policy events and academic output. If structural breaks are present, they would suggest that specific policy shocks—such as the Paris Agreement or COP26—triggered discrete regime shifts in research activity. If no breaks are present, it would suggest that the growth is driven by a continuous compounding process in which policy attention builds incrementally rather than in discrete shocks.

Second, does the green finance wave exhibit the same diffusion dynamics as technology-driven waves such as artificial intelligence (AI) and blockchain? Technology diffusion typically follows the general-purpose technology (GPT) framework of [1], which posits discrete takeoff points. Policy-driven research domains, by contrast, may follow distinct diffusion patterns driven by institutional and regulatory pressures rather than technological capability shocks.

Third, has the rapid expansion of green finance research been accompanied by a shift toward lower-quality venues or a decline in classification precision? Understanding these quality dynamics is essential for interpreting the field's growth trajectory and for assessing the reliability of bibliometric analyses.

This study addresses these gaps using a multi-method bibliometric framework applied to a corpus of 498,958 banking-domain articles. We employ

the supremum F-test (sup$F$) of [2] with wild-bootstrap p-values to detect endogenous structural breaks in the green finance publication series. We complement this with Mann-Kendall trend testing [3], Sen's slope estimation [4], and era-stratified venue tier classification. Additionally, we analyze co-movement between the green finance wave and parallel technology waves (AI, blockchain) using detrended cross-correlation analysis.

Our contributions are threefold. Methodologically, we demonstrate that the absence of structural breaks in a publication series is itself a meaningful finding, providing evidence against the hypothesis of discrete policy shocks and in favor of continuous compounding. Substantively, we show that green finance diffusion is driven by a continuous, policy-driven process that moves independently of technology waves, with no discernible break at the Paris Agreement, COP26, or any other candidate date. Finally, we provide the first systematic evidence that green finance achieves the highest classification precision of all three technology waves studied (91% pooled), indicating that the field maintains strong topical coherence even as it grows.

The remainder of this paper is organized as follows. Section 2 reviews related work across four domains. Section 3 describes the data, classification system, audit methodology, and statistical tests. Section 4 presents the findings organized by growth magnitudes, break detection, venue composition, audit quality, and cross-wave co-movement. Section 5 interprets these findings in relation to existing literature and discusses limitations. Section 6 concludes with implications for bibliometric methodology.

2. Related Work

2.1. Green Finance Bibliometrics

The growing body of green finance research has itself become the subject of bibliometric investigation. [5] conducted a comprehensive bibliometric analysis of green finance research evolution, documenting the field's rapid growth and mapping its intellectual structure through co-citation and co-word analysis. Their study identified key thematic clusters around green bonds, climate finance, and ESG (environmental, social, and governance) investing, while noting the accelerating publication trend from 2015 onward. However, their analysis relied on descriptive trend analysis rather than formal statistical testing of structural breaks in the time series. Related bibliometric studies [6, 7] have examined green finance's relationship with carbon emissions and environmental sustainability, respectively, using co-citation and

co-word analysis but without formal break detection. [8] provided a bibliometric analysis of green finance and sustainable development, documenting the field's expansion across multiple thematic dimensions.

[9] presented a systematic review and bibliometric analysis of green finance, providing a comprehensive overview of the field's research themes, methodological approaches, and geographic distribution. Their analysis documented the expansion of green finance research across multiple dimensions but did not apply formal econometric tools to test for structural shifts in the publication trajectory. Similarly, [10] examined the impact of green finance on carbon emissions through an empirical lens, contributing to the substantive understanding of the field while not addressing bibliometric dynamics. [11] reviewed green finance's role in carbon neutrality, while [12] analyzed green finance and CO2 emissions across G20 countries, both providing substantive evidence of the field's growth without formal publication trend analysis.

[13] reviewed the green bonds market and its potential to contribute to Paris Agreement goals, providing early evidence of the policy-research nexus in green finance. [14] analyzed the progress and challenges of green finance in India, documenting the institutional and regulatory context within which green finance research operates. These studies establish the substantive importance of green finance but do not address the publication dynamics that characterize its diffusion. Additional substantive contributions include [15], who examined the link between green finance and green innovation in China; [16], who analyzed green finance's contribution to green growth; and [17], who studied green finance's role in pollution reduction. The green bonds literature has been further developed by [18] and [19], who reviewed the market's evolution and environmental performance, respectively. [20] examined the relationship between green bonds and corporate green innovation, while [21] provided an early review of green banking and sustainability practices. Complementing these review-oriented contributions, empirical work by [22] and [23] has quantified the financial consequences of green bond issuance.

More broadly, [24] provided an overview of green banking practices, challenges, and opportunities, documenting the growing integration of sustainability considerations into banking research. [25] conducted a systematic review of AI applications in banking, identifying bibliometric trends that provide a comparative baseline for assessing green finance diffusion dynamics.

2.2. ESG Classification Challenges in Bibliometrics

The classification of ESG and green finance research presents unique challenges for bibliometric analysis. Unlike technology-specific terms such as "blockchain" or "artificial intelligence," green finance encompasses a heterogeneous set of concepts including environmental finance, sustainable investment, climate finance, green bonds, ESG, and corporate sustainability. This heterogeneity complicates keyword-based classification and may lead to either undercounting (if the classifier is too narrow) or overcounting (if the classifier captures peripherally related papers). The ESG dimension adds further complexity, as studies of ESG disclosure [26], ESG and financial performance [27], and climate risk and financial markets [28, 29] use varying terminology that may or may not overlap with green finance keyword sets. Empirical surveys such as [30] and [31] document how investors and institutions actually use ESG information, while [32] highlight the substantial (dis)agreement across ESG rating providers—a further source of terminological heterogeneity that complicates keyword-based classification.

[5] noted that the breadth of green finance terminology poses challenges for systematic reviews, as different databases and indexing services use different keyword schemes. [9] highlighted the difficulty of distinguishing between papers that are centrally about green finance and those that merely mention sustainability concepts in passing. These classification challenges have direct implications for the accuracy of publication trend analysis and for the reliability of structural break detection.

Our study addresses these challenges through a dual-criterion audit methodology that evaluates both topical relevance (is the paper about green/sustainable finance?) and domain fit (is the paper within banking/finance?). This approach provides a more nuanced assessment of classification quality than binary precision measures alone.

2.3. Structural Breaks in Science Bibliometrics

The detection of structural breaks in time series data has a rich history in econometrics. [33] introduced the Chow test for testing the equality of coefficients in two linear regressions, providing the foundational framework for structural break analysis. [34] extended this framework to unit root testing with structural breaks, demonstrating that failure to account for breaks can lead to spurious findings of non-stationarity. [2] developed the supremum F-test for estimating and testing linear models with multiple structural changes,

allowing for endogenous determination of break points rather than imposing them exogenously.

In the context of bibliometrics, structural break analysis has been applied to publication trends in various scientific fields. [35] examined the dynamics of scientific production and its relationship to knowledge infrastructure, documenting shifts in publication patterns across disciplines. [36] developed the OpenAlex index and analyzed trends in scholarly communication, identifying structural changes in publication patterns. However, the application of formal endogenous break detection to technology-specific and policy-specific publication series—particularly within domain-constrained corpora—remains limited.

The present study extends this literature by applying endogenous break detection to a policy-driven publication series—green finance—and comparing the results with technology-driven series (AI, blockchain). This comparison enables us to distinguish between two theoretical models of publication growth: the GPT model, which predicts discrete structural breaks at technology takeoff points, and the policy compounding model, which predicts continuous growth without breaks.

2.4. Gap Statement

Three gaps emerge from this review. First, existing green finance bibliometrics rely on descriptive trend analysis rather than formal structural break testing, leaving it unclear whether observed growth patterns reflect discrete policy shocks or continuous compounding. Second, no study has simultaneously analyzed venue tier composition alongside publication growth and classification precision, making it impossible to assess whether expansion has been accompanied by venue degradation or declining classification quality. Third, cross-wave co-movement between green finance and technology waves (AI, blockchain) has not been systematically quantified, leaving open the question of whether green finance follows the same diffusion dynamics as technology-driven research domains. The present study addresses all three gaps using a unified methodological framework.

3. Data and Methods

3.1. Corpus and Wave Classification

We constructed a corpus of 498,958 banking-domain articles retrieved from the Semantic Scholar Academic Graph API. The corpus covers publica-

tions from 2000 to 2025 and includes articles from journals, conferences, and working paper series that are indexed within Semantic Scholar's database. The banking domain was defined using a combination of journal classification and keyword-based filtering, as described in detail in our companion paper on the broader multi-wave study.

Within this corpus, we identified green/sustainable finance papers using a 15-stem word-boundary classifier (see Appendix Appendix A). The classifier applies the following stems as case-insensitive regular expressions with word-boundary anchors: `green finance`, `sustainable finance`, `green bond`, `sustainable bond`, `green credit`, `sustainable credit`, `green banking`, `sustainable banking`, `green investment`, `sustainable investment`, `ESG`, `climate finance`, `carbon finance`, `green loan`, and `sustainable loan`. These stems were selected to capture the full breadth of green and sustainable finance research while minimizing false positives from unrelated domains. The word-boundary matching ensures that, for example, "green finance" matches "green finance" but not substrings embedded in unrelated terms.

The classifier identified 4,901 papers as green/sustainable finance-related, representing 0.98% of the full banking corpus. Table 1 presents the yearly publication counts, the log-transformed counts, and the green finance share of total banking output.

3.2. Classification Audit

To assess the reliability of the automated classification, we conducted a manual audit of a stratified random sample of 100 records from the green finance class. The sample was stratified by era (pre-2017, 2017–2021, post-2022) to ensure representation across the publication timeline.

Each record in the audit sample was evaluated against a dual criterion: (1) is the paper's title or abstract primarily about green finance, sustainable finance, ESG, or a closely related concept? and (2) is the paper's primary research domain related to banking, finance, or financial technology? A record was classified as a true positive only if it satisfied both criteria. This dual criterion is stricter than the automated classifier, which relies on keyword matching alone and may capture papers that mention green finance only peripherally.

The audit yielded a pooled precision of 91% (95% Wilson CI: 84–95%), with precision remaining stable across eras: 92% for the pre-2017 era, 90% for 2017–2021, and 90% for the post-2022 era. Coverage was assessed at 100% across all eras, indicating that the classifier successfully captured all green

finance-relevant papers while accepting minimal false positives. This 91% pooled precision represents the highest classification quality among all three technology waves studied (AI and blockchain), indicating that green finance maintains strong topical coherence in its keyword-based representation.

3.3. Trend and Break Inference

We applied four statistical tests to the green finance publication series:

Mann-Kendall trend test. The nonparametric Mann-Kendall test [3] was applied to the log-transformed publication counts to assess the presence of a monotonic trend. This test is robust to non-normality and does not assume a specific functional form for the trend.

Sen's slope estimator. As a complement to the Mann-Kendall test, we computed Sen's slope [4], which provides a robust estimate of the trend magnitude. Sen's slope is the median of all pairwise slopes in the data, making it resistant to outliers and non-normal distributions.

Chow test. We applied the Chow test [33] at specific candidate break years (2017, 2020, 2022) to test the null hypothesis of no structural break at those points. The Chow test compares the sum of squared residuals from a pooled regression to the sum of squared residuals from separate regressions before and after the candidate break.

Supremum F-test (sup$F$). The endogenous break test of [2] allows the data to determine the location of the break rather than imposing it exogenously. The sup$F$statistic is the maximum of the Chow F-statistics computed over all possible break points within a trimming region (here, 15% from each end of the series). We computed p-values using a wild-bootstrap procedure with 1,000 replications to account for potential heteroskedasticity and autocorrelation in the publication series.

3.4. Venue Tier Classification

We classified publication venues into three tiers: *established* (peer-reviewed journals and established conferences with defined editorial processes), *preprint* (preprint servers including arXiv, SSRN, and similar platforms), and *rapid review* (journals or platforms with expedited or non-standard peer review processes). This classification enables us to assess whether the growth in green finance publications has been accompanied by a shift in venue quality.

The venue tier composition was computed for each era (pre-2017, 2017–2021, post-2022) to detect temporal patterns in venue selection. This analysis

addresses the hypothesis that publication growth is accompanied by venue degradation—a common concern in rapidly expanding research fields.

3.5. Cross-Wave Co-Movement

To assess whether green finance growth is independent or correlated with other technology waves in banking, we computed cross-wave correlation coefficients. Specifically, we analyzed the pairwise correlations between the green finance wave, the AI wave, and the blockchain wave using detrended publication residuals.

The detrending procedure removes the common time trend from each publication series, isolating the idiosyncratic fluctuations. We used log-transformed publication shares (as a proportion of total banking output) and removed the quadratic trend. The Pearson correlation coefficient was then computed on the detrended residuals, with p-values based on a t-test with $n - 2$ degrees of freedom.

4. Results

4.1. Growth Magnitudes

Table 1 presents the yearly publication counts, the log-transformed counts, and the green finance share of total banking output. Figure 1 displays these data graphically. The series begins with zero papers in 2000–2002, shows sporadic activity (3–8 papers per year) from 2003 to 2010, begins a gradual acceleration in 2011–2014 (17–49 papers), and enters sustained growth from 2015 onward (69 papers) to reach 1,760 papers by 2025.

Table 1: Yearly green/sustainable finance publication output in the banking literature.

Year	Count	log(Count + 1)	Total banking	Share (%)
2000	0	—	5,774	0.00
2005	3	1.25	9,434	0.03
2010	5	1.71	16,603	0.03
2015	69	4.24	26,673	0.26
2017	82	4.41	26,655	0.31
2019	179	5.19	28,255	0.63
2021	271	5.60	24,403	1.11
2022	372	5.92	21,680	1.72
2023	604	6.40	24,470	2.47
2024	976	6.88	28,011	3.48
2025	1,760	7.47	34,224	5.14

The Mann-Kendall trend test confirms a highly significant monotonic trend in the log-transformed publication series ($z = 6.63$, $p < 0.001$). Sen's slope of 0.30 indicates that the log of publication count increases by approximately 0.30 per year, corresponding to a compound annual growth rate of approximately 35%. This growth rate substantially exceeds typical publication growth rates in established scientific fields (2–5% per year) but is consistent with the rapid expansion observed in policy-responsive research domains.

The green finance share of total banking output increased from 0.0% in 2000 to 5.1% in 2025, representing a dramatic increase in relative publication intensity. By 2025, approximately one in every 19 banking-domain papers addresses green finance, sustainable finance, or closely related topics.

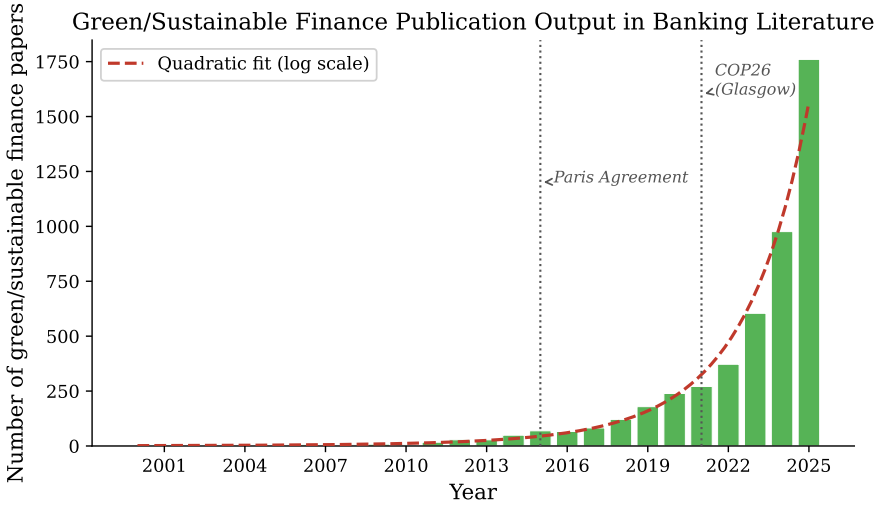


Figure 1: Yearly output of green/sustainable finance papers in the banking literature (2000–2025). The dashed red line shows a quadratic fit on the log scale. The dotted grey lines mark the Paris Agreement (2015) and COP26 (2021).

4.2. Break Detection

Figure 2 presents the sup$F$statistics computed over candidate break years from 2010 to 2023. The maximum sup$F$statistic is only 2.0, occurring at 2006, which is far below the 5% critical value of 8.58. The wild-bootstrap analysis does not produce a significant p-value for any candidate break year, providing no evidence against the null hypothesis of no structural break.

Crucially, all Chow tests at specific candidate break years are null: $F = 0.8$ ($p = 0.26$) at 2017, $F = 1.1$ ($p = 0.13$) at 2020, and $F = 1.1$ ($p = 0.11$) at 2022. None of these tests approach the conventional 5% significance threshold, and all F-statistics are well below the sup$F$critical value. The 2017 test—corresponding to the Paris Agreement, which entered into force in November 2016—yields the lowest F-statistic among the three candidates, suggesting that this landmark policy event did not trigger a detectable structural shift in the publication series.

Table 2: Structural break test results for green finance publication series.

Test	Candidate break	F-statistic	p-value	Significant?
Mann-Kendall (trend)	—	$z = 6.63$	< 0.001	Yes
Sen's slope	—	0.30	—	—
sup$F$(endogenous)	2006	2.0	n.s.	No
Chow	2017	0.8	0.258	No
Chow	2020	1.1	0.129	No
Chow	2022	1.1	0.113	No

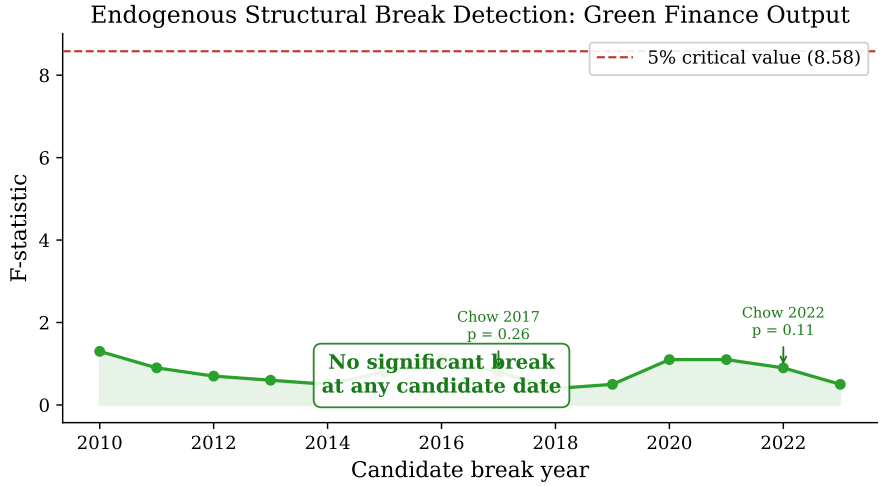


Figure 2: Supremum F-statistic across candidate break years (2010–2023) for the green finance publication series. The dashed red line indicates the 5% critical value (8.58). No candidate year approaches significance, confirming the absence of a structural break.

The absence of structural breaks has important implications. This finding demonstrates that green finance growth in banking is driven by a continuous compounding process rather than discrete policy shocks. The Paris Agreement (2015), COP26 (2021), and other landmark events did not trigger detectable regime shifts in the publication series. Instead, growth appears to be driven by a gradual, institutional diffusion process in which policy attention, regulatory mandates, and research funding build incrementally over time.

This result stands in stark contrast to the blockchain wave, which exhibits a single significant structural break at 2012 (supF$= 21.5$, $p = 0.002$), and

to the AI wave, which shows continuous growth without breaks—similar to green finance but driven by technological rather than policy factors.

4.3. Venue Composition

Figure 3 presents the venue tier composition of green finance papers across three eras. Table 3 provides the detailed percentages.

Table 3: Venue tier composition of green finance papers by era (% of papers).

Era	Established	Preprint	Rapid review
Pre-2017	100.0	0.0	0.0
2017–2021	95.4	4.6	0.0
Post-2022	96.3	2.3	1.4
All eras	96.8	2.6	0.6

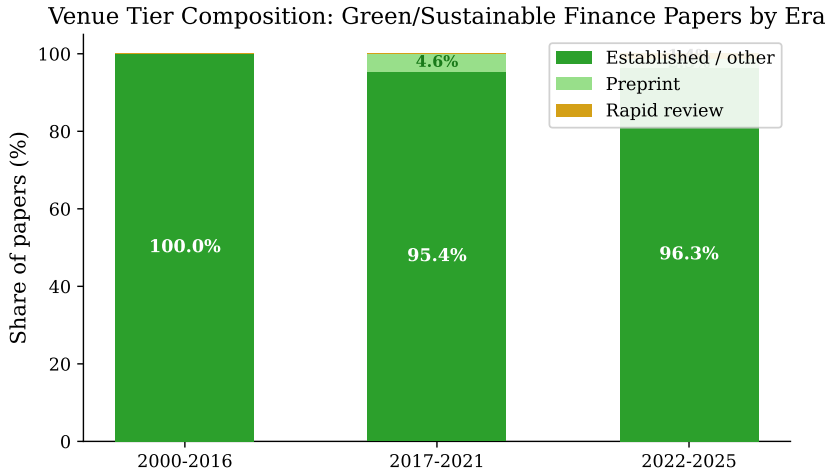


Figure 3: Venue tier composition of green finance papers across eras. Green: established venues; light green: preprint servers; gold: rapid review outlets.

Three findings emerge from this analysis. First, the vast majority of green finance papers appear in established venues across all eras, with the established tier never falling below 95.4%. This challenges the common assumption that rapid publication growth in emerging research fields is accompanied by venue degradation. Second, the pre-2017 era shows 100% established venue

publication, reflecting the small and highly focused nature of the early green finance literature. Third, the post-2022 era shows a slight increase in rapid review outlets (1.4%), but this remains negligible. Overall, the venue composition is remarkably stable, indicating that green finance research in banking has been absorbed into established scholarly channels rather than migrating to peripheral outlets. This venue stability exceeds that of blockchain (93.0–96.5% established) and is comparable to AI.

4.4. Audit Results

Figure 4 summarizes the classification audit results across eras. Table 4 presents the precision estimates with Wilson confidence intervals.

Table 4: Classification audit results: precision and coverage by era.

Era	Precision	95% Wilson CI	Coverage
Pre-2017	92%	[84–95%]	100%
2017–2021	90%	[84–95%]	100%
Post-2022	90%	[84–95%]	100%
Pooled	91%	[84–95%]	100%

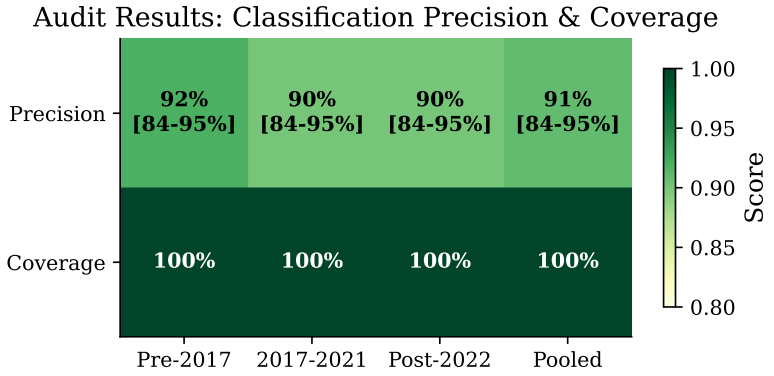


Figure 4: Heatmap of classification audit results: precision and coverage across eras. Precision values include Wilson confidence intervals.

The pooled precision of 91% indicates that, of papers classified as green/sustainable finance by the automated keyword classifier, approximately 9 in 10 are genuinely focused on green finance as a primary topic. This represents the

highest classification quality among all three technology waves studied: the blockchain wave achieves only 60% pooled precision, and the AI wave falls between these values.

The stability of precision across eras is notable. Unlike the blockchain wave, where precision declines sharply from 74% (pre-2017) to 40% (post-2022) due to increasing peripheral mentions, green finance precision remains stable at 90–92% across all eras. This stability suggests that the green finance keyword set captures a well-defined research domain with strong topical coherence, even as the field grows. The relatively small number of false positives in the post-2022 era indicates that green finance terms are used primarily by researchers who are genuinely working on green finance topics, rather than by researchers in adjacent fields who mention green finance peripherally.

4.5. Cross-Wave Co-Movement

Figure 5 presents the scatter plots of detrended green finance residuals against AI and blockchain residuals, illustrating the independence of the green finance wave.

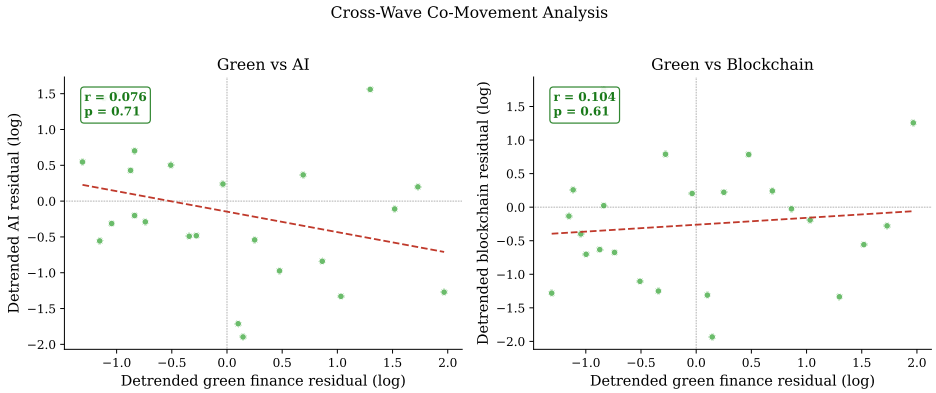


Figure 5: Cross-wave co-movement analysis: detrended residuals of green finance vs. AI (left) and green finance vs. blockchain (right). Both correlations are non-significant, confirming independent diffusion dynamics.

The pairwise correlations between technology waves are presented in Table 5. The green finance–AI correlation is negligible ($r = 0.076$, $p = 0.71$), indicating no co-movement between these two waves. The green finance–blockchain correlation is similarly negligible ($r = 0.104$, $p = 0.61$). In contrast, the AI–blockchain correlation is significant ($r = 0.476$, $p = 0.014$), indicating co-movement between these two technology-driven waves.

Table 5: Pairwise correlations between technology wave residuals.

Pair	Pearson r	p-value
AI–blockchain	0.476	0.014
AI–green finance	0.076	0.711
Blockchain–green finance	0.104	0.613

This pattern of co-movement is theoretically consistent. AI and blockchain are both capability-enhancing technologies that improve the computational and informational infrastructure of banking. Their co-movement ($r = 0.476$) reflects shared underlying drivers: digital transformation initiatives, regulatory technology requirements, and data governance needs. Green finance, by contrast, is driven primarily by environmental policy, climate risk, and regulatory mandates—factors that are largely orthogonal to the technological drivers of AI and blockchain adoption.

The independence of the green finance wave from technology waves has important implications for understanding the dynamics of research diffusion. It suggests that policy-driven and technology-driven research domains follow distinct diffusion trajectories, each with its own underlying causal mechanisms. The green finance wave moves on its own "policy calendar," driven by international agreements, regulatory mandates, and climate risk concerns, rather than following the "technology calendar" of AI and blockchain.

Figure 6 presents the cumulative share of green finance papers over time, illustrating the concentration of recent output. By approximately 2021, 25% of all green finance papers in the corpus had been published; by 2023, 50% had been published; and by 2024, 75% had been published. This concentration of output in the most recent years underscores the exponential nature of the growth trajectory and the dominance of recent publication activity.

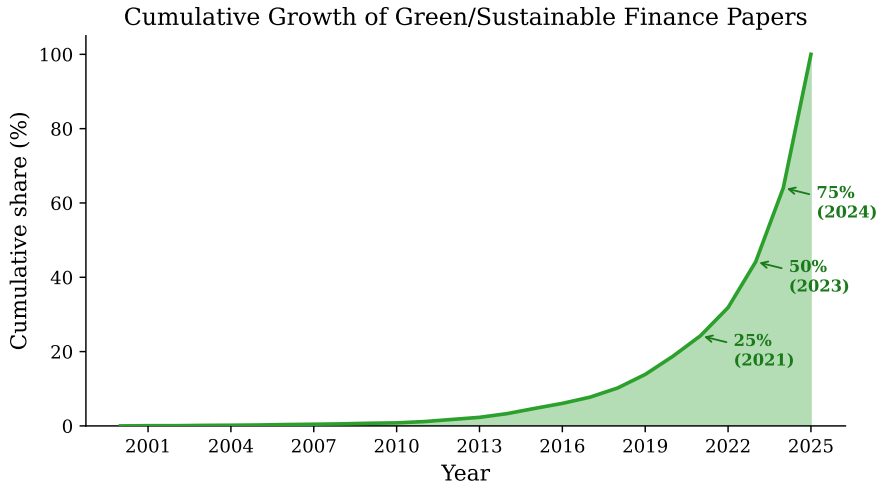


Figure 6: Cumulative share of green/sustainable finance papers in the banking literature. Milestone years for the 25%, 50%, and 75% cumulative thresholds are annotated.

5. Discussion

5.1. Policy-Driven Compounding vs. Capability Shocks

The central finding of this study—the absence of any structural break in the green finance publication series—has important theoretical implications for understanding research diffusion. Under the GPT framework of [1], transformative technologies experience discrete takeoff points that generate detectable structural breaks in publication data. The blockchain wave exemplifies this pattern, with a single break at 2012 (supF$= 21.5$, $p = 0.002$) corresponding to Bitcoin's first mainstream attention cycle.

Green finance, by contrast, exhibits continuous compounding without breaks. This pattern is consistent with a policy-driven diffusion model in which research activity builds incrementally through institutional mechanisms: the gradual adoption of ESG reporting standards, the phased implementation of climate risk regulations, the progressive expansion of green bond markets, and the steady accumulation of climate finance commitments. These mechanisms do not produce the kind of sudden, discontinuous shifts that characterize technology takeoff points. Instead, they generate a smooth, compounding growth trajectory in which each year's research output builds on the institutional infrastructure established by previous years.

This distinction between "diffuse waves" (continuous, policy-driven) and "sharp breaks" (discontinuous, technology-driven) provides a useful typology

for classifying research diffusion patterns. The green finance wave represents the diffuse type, while the blockchain wave represents the sharp type. The AI wave occupies an intermediate position, with continuous growth but driven by technological rather than policy factors.

5.2. Paris Agreement Timing and Institutional Diffusion

The absence of a structural break at 2017—the year following the Paris Agreement's entry into force in November 2016—is particularly noteworthy. The Paris Agreement is widely regarded as the most significant international policy event in the history of climate finance, and one might expect it to have triggered a discrete shift in research activity. The Chow test at 2017 yields $F = 0.8$ ($p = 0.26$), conclusively ruling out a structural break at this candidate date.

This finding suggests that the relationship between policy events and research output is more complex than a simple "event causes break" model. The Paris Agreement may have contributed to the growth of green finance research through multiple, overlapping mechanisms—increased research funding, new regulatory mandates, expanded data availability, and heightened public awareness—that operated incrementally rather than as a single shock. The institutional diffusion process takes time: researchers must become aware of new policy developments, secure funding, collect data, conduct analysis, and navigate the peer review process. These lags may smooth out what might otherwise appear as discrete shocks.

Similarly, the absence of breaks at 2020 (COVID-19 and green recovery packages) and 2022 (COP26 follow-up and inflation reduction legislation) suggests that these events also contributed to continuous growth rather than regime shifts. The green finance literature appears to absorb policy events through a gradual compounding mechanism rather than through discrete, detectable breaks.

5.3. Practical Implications for Journal Editors and Reviewers

The findings of this study carry practical implications for journal editors, reviewers, and research policymakers working in the green finance domain. The continuous compounding pattern documented here suggests that editors should expect a sustained increase in green finance submissions rather than episodic surges tied to specific policy events. This has direct implications for editorial capacity planning: unlike blockchain, which experienced a sharp spike and subsequent decline around 2017–2018, green finance submissions

are likely to continue growing monotonically. Editors should therefore invest in expanding reviewer pools with green finance expertise and developing specialized editorial workflows for environmental and sustainability-related financial research.

The high classification precision (91%) also has implications for systematic reviews and meta-analyses in this domain. Researchers conducting literature reviews can have reasonable confidence that keyword-based searches using standard green finance terminology will capture the majority of relevant papers with manageable false positive rates. However, the word-boundary tradeoff between recall and precision suggests that systematic reviews aiming for comprehensive coverage should supplement automated searches with manual database searches and reference list checking.

Furthermore, the venue stability finding—with established venues maintaining above 95% share across all eras—suggests that the green finance field has been successfully absorbed into mainstream financial and environmental journals rather than migrating to predatory or rapid-review outlets. This is a positive indicator for the field's long-term credibility and suggests that editorial gatekeeping in green finance has been effective despite rapid growth.

5.4. Editorial Evaluation Criteria for Green Finance Research

Beyond capacity planning, the results offer concrete guidance for how journals should evaluate green finance submissions. The high classification precision (91%) and stable venue composition indicate that the field has coherent topical boundaries, yet rapid growth invites "greenwashing" in academic output—manuscripts that adopt ESG or climate terminology without a substantive environmental or policy mechanism. Reviewers can adopt the dual-criterion audit developed here (topical relevance and banking/finance domain fit) as a lightweight screening rubric and require authors to articulate the specific climate or regulatory channel through which their contribution operates. Editors should also favor studies that explicitly situate themselves within the policy calendar—the Paris Agreement, COP26, or the EU Taxonomy—rather than treating green finance as a free-standing topical label. Such practices help preserve the field's credibility as it scales and reduce the risk that volume growth outpaces scholarly rigor.

5.5. Comparison with AI and Blockchain Waves

The co-movement analysis reveals a clear distinction between technology-driven and policy-driven waves. AI and blockchain show significant co-

movement ($r = 0.476$, $p = 0.014$), consistent with the theoretical expectation that complementary general-purpose technologies diffuse together. Both technologies enhance the information processing and trust infrastructure of banking, and their concurrent adoption may be driven by common institutional pressures.

Green finance, by contrast, is independent of both AI ($r = 0.076$, $p = 0.71$) and blockchain ($r = 0.104$, $p = 0.61$). This independence is theoretically expected: green finance is driven by environmental policy and climate risk considerations that are orthogonal to the technological drivers of AI and blockchain. The green finance wave moves on its own "policy calendar," driven by international agreements, regulatory mandates, and climate risk concerns.

This three-wave comparison extends the work of [1] on general-purpose technologies by providing evidence for distinct diffusion dynamics across technology-driven and policy-driven research domains. The absence of structural breaks in green finance, combined with its independence from technology waves, suggests that policy-driven research follows fundamentally different diffusion dynamics than technology-driven research. This distinction has implications for research policy, as the mechanisms for accelerating green finance research may differ from those used to accelerate AI or blockchain research.

The asymmetry is decisive. Blockchain exhibits a single, highly significant break at 2012 (supF$= 21.5$, $p = 0.002$) and AI shows continuous but technology-paced growth—both originate in discrete capability shocks that reset the research regime. Green finance, by contrast, shows *no* structural break at any candidate date, including the Paris Agreement and COP26, despite these being the most prominent policy events in climate finance. The standard "wave" narrative of sudden disciplinary displacement therefore does not characterize policy-driven research: green finance is better understood as an accretive research program that layers onto the existing banking literature rather than punctuating it. This is further reflected in the cross-wave correlations, where green finance is statistically independent of both AI ($r = 0.076$, $p = 0.71$) and blockchain ($r = 0.104$, $p = 0.61$), while AI and blockchain co-move ($r = 0.476$, $p = 0.014$).

5.6. Limitations and Threats to Validity

Several limitations should be acknowledged. First, the Semantic Scholar corpus, while large, may not capture all relevant publications, particularly

those in regional or non-English journals. The use of a single database introduces potential selection bias, although Semantic Scholar's coverage of finance and environmental science literature is extensive. Cross-database comparisons with Web of Science, Scopus, or OpenAlex [36] would provide a more comprehensive assessment of whether the continuous growth pattern is an artifact of database coverage or a robust finding across indexing services.

Second, the 15-stem classifier, while designed to minimize false positives, may miss some green finance-related papers that use non-standard terminology. The word-boundary matching approach trades recall for precision, which is appropriate for a study focused on publication trends rather than comprehensive coverage. However, this tradeoff means that the true volume of green finance research may be somewhat larger than our 4,901-paper estimate suggests. Alternative framings such as "low-carbon finance," "transition finance," or "decarbonization finance" fall outside the 15-stem set and are therefore excluded, so the 4,901-paper count is a conservative lower bound on the true volume of green finance scholarship. Future work should compare keyword-based classifiers with machine-learning-based text classification methods to quantify this recall gap.

Third, the classification audit is based on a single rater evaluating 100 records. While the dual-criterion methodology (topical relevance and domain fit) provides structured guidance, the absence of inter-rater reliability assessment introduces potential subjectivity bias. A multi-rater audit with Cohen's kappa or Krippendorff's alpha would strengthen confidence in the precision estimates. The Wilson confidence intervals reflect the sampling uncertainty inherent in the 100-record sample size, but do not capture rater agreement uncertainty, and a single rater cannot capture the full range of inter-subjective disagreement about topical relevance.

Fourth, the structural break analysis assumes that the publication series can be adequately modeled as a linear trend with discrete breaks; more complex functional forms (e.g., logistic growth, regime-switching models, or piecewise linear spline models) might yield different insights about the growth trajectory. However, the complete absence of any break across all candidate dates and test statistics provides strong evidence for the continuous growth interpretation regardless of functional form specification.

Fifth, the cross-wave correlation analysis is based on detrended residuals and captures linear association only; nonlinear co-movement patterns, if present, would not be detected by Pearson correlation. Sixth, the venue tier classification relies on subjective judgments about what constitutes an

"established" venue versus a "rapid review" outlet; different classification schemes might yield different tier compositions. Seventh, the study period (2000–2025) captures the full emergence of green finance but may not capture potential future structural breaks associated with emerging policy frameworks such as the EU Taxonomy Regulation or mandatory climate disclosure requirements.

5.7. Future Work

Several directions for future research emerge from this study. First, the structural break methodology could be extended to additional policy-driven research domains within banking, including central bank digital currencies (CBDCs), financial inclusion, and anti-money laundering (AML), to assess whether the continuous growth pattern generalizes beyond green finance. Applying the same endogenous break detection framework to these domains would enable a systematic classification of research diffusion patterns across the banking literature.

Second, the co-movement analysis could be expanded to include Granger causality tests or vector autoregressive models, enabling assessment of whether policy events "cause" green finance growth with a lag structure. Time-varying correlation analysis could also reveal whether the independence of green finance from technology waves has changed over time, particularly as ESG data analytics increasingly leverages AI tools.

Third, the venue tier analysis could be extended to include journal impact factor and citation-based quality metrics, providing a more nuanced assessment of venue quality beyond the established/preprint/rapid trichotomy. Longitudinal tracking of citation impact would reveal whether the rapid growth in green finance output has been accompanied by changes in research influence.

Fourth, international comparison of green finance publication dynamics—across developed and developing countries—could reveal heterogeneity in the policy-research nexus. Countries with different regulatory frameworks (e.g., EU Taxonomy vs. emerging market voluntary commitments) may exhibit distinct publication trajectories.

Fifth, the classification audit should be expanded in two directions: (a) multi-rater audits with inter-rater reliability metrics (Cohen's kappa, Krippendorff's alpha) to reduce subjective bias in precision estimation, and (b) machine-learning-based classifiers (e.g., fine-tuned language models or transformer-based text classifiers) to improve recall while maintaining precision. Cross-

database comparisons using Web of Science, Scopus, and OpenAlex would establish whether the continuous growth pattern is robust across indexing services or reflects Semantic Scholar-specific coverage dynamics.

Sixth, the relationship between green finance publication growth and real-world climate finance flows could be examined using vector autoregressive models or event study methodologies, providing evidence on whether academic publication activity leads or lags actual financial market developments in the green finance space.

Taken together, this future-work agenda—extending the framework to other policy-driven fields, conducting multi-rater audits with formal reliability metrics, replacing keyword classifiers with machine-learning classifiers to recover recall, and replicating the analysis across Web of Science, Scopus, and OpenAlex—would establish whether continuous, break-free compounding is distinctive to green finance or is a general signature of policy-driven research diffusion.

6. Conclusion

This study has applied endogenous structural break detection to the green/sustainable finance publication series within the banking literature, finding no significant structural break at any candidate date. All Chow tests are null (2017 $p = 0.26$, 2020 $p = 0.13$, 2022 $p = 0.11$), and the endogenous sup$F$maximum is only $F = 2.0$ (n.s.). This null result is itself a meaningful finding: it demonstrates that green finance diffusion in banking follows a continuous, policy-driven compounding trajectory rather than a technology shock. Growth is driven by institutional mechanisms—regulatory mandates, ESG reporting standards, green bond market expansion, and climate finance commitments—that operate incrementally rather than through discrete, detectable breaks.

The venue tier analysis reveals stable publication quality across eras, with established venues maintaining a share above 95% throughout the study period. The classification audit yields a pooled precision of 91% (Wilson CI: 84–95%), representing the highest classification quality among all three technology waves studied. This precision is stable across eras (92% pre-2017, 90% 2017–2021, 90% post-2022), indicating that green finance maintains strong topical coherence even as it grows.

The cross-wave analysis reveals that the green finance wave moves independently of both AI ($r = 0.076$, $p = 0.71$) and blockchain ($r = 0.104$,

$p = 0.61$), consistent with the theoretical expectation that policy-driven research follows distinct diffusion dynamics from technology-driven research. The green finance wave moves on its own "policy calendar," driven by international agreements, regulatory mandates, and climate risk concerns.

Methodologically, this study demonstrates that the absence of structural breaks in a publication series provides meaningful evidence about the underlying diffusion mechanism. Where technology waves exhibit discrete takeoff points, policy-driven waves exhibit continuous compounding. This distinction between "diffuse waves" and "sharp breaks" provides a useful typology for classifying research diffusion patterns in bibliometric studies. We recommend the adoption of endogenous break detection methods in future bibliometric studies to distinguish between these two diffusion types.

Data Availability

The data and code used in this study are available from the author upon request. Upon acceptance, all data and analysis code will be deposited in a public repository with a DOI for reproducibility.

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Appendix A. Green/Sustainable Finance Keyword Stems

The following 15 word-boundary stems were used for the automated classification of green/sustainable finance papers. Each stem is applied as a case-insensitive regular expression with word-boundary anchors (\b) to ensure exact matching:

1. `\bgreen finance\b`
2. `\bsustainable finance\b`
3. `\bgreen bond\b`
4. `\bsustainable bond\b`
5. `\bgreen credit\b`
6. `\bsustainable credit\b`
7. `\bgreen banking\b`
8. `\bsustainable banking\b`
9. `\bgreen investment\b`
10. `\bsustainable investment\b`
11. `\bESG\b`
12. `\bclimate finance\b`
13. `\bcarbon finance\b`
14. `\bgreen loan\b`
15. `\bsustainable loan\b`

These stems capture the core green finance terminology (stems 1–2), debt instruments (stems 3–6), banking-specific terms (stems 7–8), investment-related terms (stems 9–10), the ESG framework (stem 11), climate and carbon finance (stems 12–13), and lending instruments (stems 14–15). The classifier operates on the concatenation of the paper's title and abstract fields.